UNITED INTERNATIONAL UNIVERSITY (UIU)
Dept. of Computer Science & Engineering
Trimester: Fall 2026 (Project Idea Presentation)
Course No: DS 4889 **Title:** Machine Learning
Course Teacher: KFT | Kazi Fahim Tahmid
Section: BA **Deadline:** 12 November, 2026

Assalamualikum everyone,

Instructions for following "Instructions":

- Make sure you have submitted your group members' names, roll numbers, and initial research topic choices via the form link shared previously.
- Kindly, **don't use LLMs or Claude to generate your pptx**. I will instantly understand if you made it or any LLM made it. Any use of LLMs for preparation of the slides will result in instant disqualification.
- Questions will be asked on whatever you have inside your presentation slides. I recommend understanding every single word that you write in your slide. If you can't explain something that you've written in your proposal slide, it means you haven't reasoned or written those things yourself which is equivalent to cheating.
- Taking help of LLM is completely fine so long as you are understanding what's happening yourselves.
- Keep the presentation slides CLEAN. In an academic presentation slide, nothing flashy or too much designed slides are deemed as "distracting". Instead keep it simple, every slide should contain the title of the current section and a page number in the bottom is a must. In the middle potion, just include what you want to present.
- Make sure you're only keeping the key points as bullet points mostly instead of writing a whole paragraph for what you're presenting. The slide should contain just the points and you, yourself, are going to speak in your own words of what you want to mean.

Following up on the announcement regarding the upcoming Research Idea Presentation, here is a recommended slide outline to help your team structure your presentation effectively:

- **Title & Team Details**
  - Tentative research title / project idea name
  - Full names and roll numbers of all group members
  - Course section and lab/lecture details
- **Problem Statement & Motivation**
  - What specific problem, domain bottleneck, or theoretical challenge does your research address?
  - Why does this problem matter, and what prospect does it bring to the advancement or improvement of the field?

- **Literature Review & Gap Analysis**
    - Critical summary of recent, state-of-the-art works and existing methods in this domain
    - Specific limitations, unanswered questions, or performance gaps in current research (e.g., high computational cost, poor generalizability, data scarcity, lack of interpretability)

- **Proposed Research Focus & Core Concept**
    - The exact research gap or open question your team aims to address
    - Abstract/high-level overview of your proposed idea or algorithmic approach to bridge this gap

- **Methodology & Data Infrastructure (Core Focus)**
    - Specific Machine Learning paradigms, model architectures, or algorithms you plan to explore or adapt (e.g., Transformers, GNNs, Ensemble methods, Deep Reinforcement Learning, etc.)
    - Datasets, benchmark repositories, or data collection strategies you plan to use
    - Existing baseline models you plan to compare your approach against

- **Expected Impact & Validation Strategy**
    - Quantitative evaluation metrics you will use to measure performance (e.g., Precision/Recall, F1-score, MSE, AUC-ROC)
    - Anticipated contributions to the field and potential technical challenges/risks

A Few Tips for Your Presentation:

- **Ground your work in literature:** Ensure your proposed idea directly responds to findings and shortcomings highlighted in recent research papers.

- **Focus on the research gap:** Clearly articulate *why* current methods fall short and *how* your approach plans to target that limitation.

- **Use visual diagrams over text:** Pipeline schematics, gap-mapping charts, and model architecture diagrams communicate complex machine learning concepts far better than long text paragraphs.

- **Time Management:** Aim for a 5 to 7-minute presentation to allow adequate time for peer discussion and the Q&A session.

Good luck with your preparations!

